IDEATION PHASE

Define the Problem Statements

Date	25 October 2026
Team ID	Individual project
Project Name	HotelPro (Hotel Management System)
Maximum Marks	2 Marks

Customer problem statement. To define the core problems that guests, front-desk staff and hotel owners face with manual or disconnected hotel operations, and to use those insights to guide the development of HotelPro, a secure and scalable hotel management system.

Reference: miro.com/templates/customer-problem-statement

PS no.	I am (customer)	I am trying to	But	Because	Which makes me feel
PS-1	a guest planning a stay	find a room that is free on my dates and book it quickly	I have to call or message the hotel and wait for a reply	availability is tracked on paper or in separate spreadsheets	frustrated and unsure my booking is confirmed
PS-2	a front-desk staff member	check guests in and out and see who is arriving today	I cannot tell which rooms are free or occupied without checking several records	bookings, rooms and payments are not in one system	stressed and worried about double bookings
PS-3	a hotel owner or admin	know my occupancy and revenue and manage the room inventory	reports are compiled by hand at the end of the day or week	there is no single, secure place for guest, booking and payment data	uninformed and concerned about data privacy
PS-4	a guest who has already booked	pay, review or cancel my booking without visiting the hotel	there is no self-service option and refunds are unclear	the hotel has no online booking history or payment status	anxious about losing my money

Summary. The four problem statements point to one need: a single secure platform where guests book and pay online, staff run check-in and check-out from live availability, and owners see occupancy and revenue.

IDEATION PHASE

Empathize and Discover: Empathy Map Canvas

Date	25 October 2026
Team ID	Individual project
Project Name	HotelPro (Hotel Management System)
Maximum Marks	4 Marks

Empathy map canvas. Understand the needs, frustrations and behaviours of the two main users of HotelPro: the guest who books a room, and the front-desk staff member who runs the stay.

Reference: mural.co/templates/empathy-map-canvas

Persona 1: The guest

Guest planning a 2 to 3 night stay	
SAYS "Is a room free from the 3rd to the 5th?" "Can I pay online?" "How do I cancel if my plans change?"	**THINKS** Will my booking really be held? Is this price final? Is my payment information safe?
DOES Compares hotels online, checks dates and price, books through the site or by phone, keeps a screenshot as proof.	**FEELS** Hopeful when options appear, anxious while waiting for confirmation, relieved when the booking shows as confirmed.
PAINS Waiting for replies. Unclear availability. No view of booking status or payment. Fear of overbooking.	**GAINS** Instant availability for chosen dates, clear total price, booking history, easy cancellation, secure login.

Persona 2: The front-desk staff member

Front-desk staff handling arrivals and departures	
SAYS "Which rooms are free right now?" "Has this guest paid?" "Who is leaving today?"	**THINKS** I must not give the same room to two guests. I need to finish check-in quickly while a queue builds.
DOES Looks up bookings, checks guests in and out, records cash payments, answers calls about availability.	**FEELS** Pressured at peak hours, careful about mistakes, confident when the screen shows the true status.
PAINS Records spread over several places. Manual updates. Errors in room status and payment status.	**GAINS** One live list of bookings with status filters, one-click check-in and check-out, payment status at a glance.

IDEATION PHASE

Brainstorm and Idea Prioritization

Date	25 October 2026
Team ID	Individual project
Project Name	HotelPro (Hotel Management System)
Maximum Marks	4 Marks

Brainstorming gives an open space to generate many ideas first and judge them afterwards. Because HotelPro is an individual project, the session was run by one person, using the problem statements PS-1 to PS-4 as the starting point.

Reference: mural.co/templates/brainstorm-and-idea-prioritization

Step 1: Team gathering, collaboration and selecting the problem statement

- Team: one member (individual project), acting as developer, designer and tester.
- Selected problem: PS-1 to PS-4 combined: *"How might we let guests book and pay online while giving staff and owners live control of rooms, bookings and revenue?"*

Step 2: Brainstorm, idea listing and grouping

Group	Ideas
Booking and availability	Search by dates, guests and room type. Show only rooms free on every night. Block double bookings. Live price calculation. Guest booking history and cancellation.
Front-desk operations	Check-in and check-out workflow. Filter bookings by status. Mark cash payments. Room maintenance status. Housekeeping task board.
Payments	Online payment (UPI, card, net banking) with a transaction reference. Pay at hotel. Refund status on cancellation. Real gateway integration.
Security and administration	Registration and login with hashed passwords. JWT sessions. Roles: guest, staff, admin. Room inventory management. Rate limiting and input validation.
Insight and growth	Dashboard with occupancy and revenue. Guest list. Email or SMS confirmations. Reviews and ratings. Photo gallery. Multi-property support. Dynamic pricing.

Step 3: Idea prioritization (impact against effort)

All high-impact ideas with low or medium effort form the first release. Ideas that need external services or large extra effort are kept for later versions.

Idea	Impact	Effort	Decision
Search rooms by dates, guests and type	High	Medium	MVP (build now)
Prevent double bookings	High	Medium	MVP (build now)
Book a room with live price	High	Low	MVP (build now)
Registration, login and role-based access	High	Medium	MVP (build now)
Room inventory management (admin)	High	Low	MVP (build now)
Check-in and check-out workflow	High	Low	MVP (build now)
Guest booking history and cancellation	Medium	Low	MVP (build now)
Simulated online payment and pay at hotel	Medium	Medium	MVP (build now)
Dashboard: occupancy, arrivals, revenue	Medium	Medium	MVP (build now)
Real payment gateway (Razorpay or Stripe)	High	High	Later
Email or SMS confirmations	Medium	Medium	Later
Room photo gallery, reviews and ratings	Medium	Medium	Later
Housekeeping task board	Medium	High	Later
Multi-property support and dynamic pricing	Medium	High	Later